C Flexible Automation (Soft Automation)

It has the capability of producing a variety of parts with minimal change over time from one part to the next.

Advantages

- Customized products
- Design variations can be dealt with flexibility
- Parts with complex shapes can be produced.

Disadvantages

- High initial investment
- Unit cost is relatively high

10.3 Needs for Industrial Automation

- To increase the labour productivity
- To improve the product quality
- To reduce the labour or production cost
- To reduce routine manual tasks
- To improve safety
- To assist remote monitoring

10.4 The advantages of automation

- Increased throughout productivity.
- Improved quality or increased predictability of quality
- Reduce overall production cost
- Less floor area required
- Human fatigue is greatly minimized
- Replaces humans in tasks done in dangerous environments (i.e. fire, space, volcanoes, nuclear facilities, underwater, etc.)
- Reduces operation time and work handling time significantly.
- Reduce maintenance requirements
- Uniform components are produced

The disadvantages of automation

- Possible security threats/vulnerability due to increased relative susceptibility for committing errors.
- Unpredictable or excessive development costs.
- High initial cost.
- Displaces workers due to job replacement.
- Leads to further environmental damage and could compound climate change.

10.5 Applications

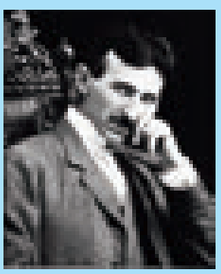
- Numerical Control Machines
- Automated production lines
- Automated assemble lines
- Robots in manufacturing
- Flexible manufacturing system
- Automation in Daily life

உங்களுக்குத் தெரியுமா?

Father of automation

Nikola Tesla (1856 – 1943)

"Nikola Tesla" is called the father of automation. Year: 1856 – 1943. He was a Serbian – American inventor, Electrical engineer, Mechanical Engineer.



Nikola Tesla
(1856 – 1943)

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